

CERTICUT Conceptual Model

Common Logical Problem

Same source circuit, logical wires, fragment capacities, and feasible placements

Representation a

Gate decomposition
QPD interaction weights $w^{(a)}$

Representation b

Equivalent decomposition
QPD interaction weights $w^{(b)}$

Shared Placement Decision Space

Optimize each weighted graph over the same feasible partitions; compare their optimal-placement sets tie-safely

Part I: Anytime Overhead Bounds

Global MIP bounds quantify independent-QPD model overhead

Part II: Placement Stability

Regret quantifies changed decisions; margin conditions guarantee stability